

Lecture 8: Behavioural Finance

Bubbles, Crashes, Herding, and the Limits of Arbitrage

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Opening — The Cisco Impossibility

March 2000: Cisco Systems Hits \$555 Billion Market Cap

To justify this price using standard discounted cash flow analysis:

- Cisco needed to grow at **80% annually for 10 years**.
- Then maintain **20% annual growth** indefinitely.
- This would make Cisco **larger than the entire US economy**.

Did investors know this was impossible? Yes. Did they invest anyway? Yes.

The Dot-Com Bubble 2000

- NASDAQ rose **527%** from 1995 to March 2000.
- NASDAQ then fell **77%** from March 2000 to October 2002.
- This is what behavioural finance studies: why rational-looking investors create irrational markets.

1. **Efficient Market Hypothesis** — benchmark and evidence for it
2. **Anomalies** — evidence against EMH
3. **Limits of Arbitrage** — why smart money can't always win
4. **Noise Traders** — the model (DeLong et al., 1990)
5. **Asset Bubbles** — anatomy and real examples
6. **Herding and Cascades** — when rational agents create irrational markets
7. **Market Anomalies** — momentum, disposition, home bias
8. **2008 Financial Crisis** — a behavioural post-mortem
9. **Canadian housing bubble (2020–2022)** — close to home

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The Efficient Market Hypothesis (Fama, 1970)

All available information is instantly reflected in prices. No strategy can systematically generate **risk-adjusted abnormal returns**.

Three Forms:

1. **Weak:** Past prices don't predict future prices.
2. **Semi-strong:** All public info reflected.
3. **Strong:** All info (including insider) reflected.

Evidence FOR: 91% of Canadian active equity funds underperformed over 15 years (SPIVA, 2023). On average, managers *must* earn market return (Sharpe, 1991).

But Anomalies Exist

Momentum, value premium, disposition effect, closed-end fund puzzle, post-earnings drift — all *systematic* patterns contradicting strict EMH.

Noise Trader Model (DeLong et al., 1990)

Two types: **rational arbitrageurs** (know fundamentals) and **noise traders** (trade on sentiment).

The Surprising Conclusion

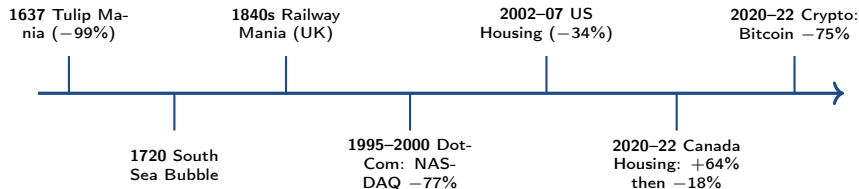
Noise traders can earn **higher expected returns** because they bear sentiment risk. Rational traders don't fully correct mispricing — “markets can stay irrational longer than you can stay solvent.”

Limits of Arbitrage (Shleifer & Vishny, 1997)

Arbitrage is risky because:

1. No perfect substitute exists (fundamental risk).
2. Mispricing can deepen before correcting (noise trader risk).
3. Arbitrageurs have short horizons and career concerns.

Historical Bubbles — A Timeline

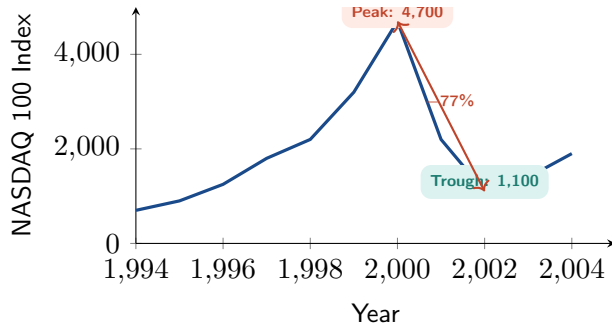


Isaac Newton on Tulips (Paraphrased)

Newton lost £20,000 in the South Sea Bubble (equivalent to $\approx$ £3M today): *“I can calculate the motion of heavenly bodies but not the madness of people.”*

The Dot-Com Bubble

NASDAQ 100: 1994–2004

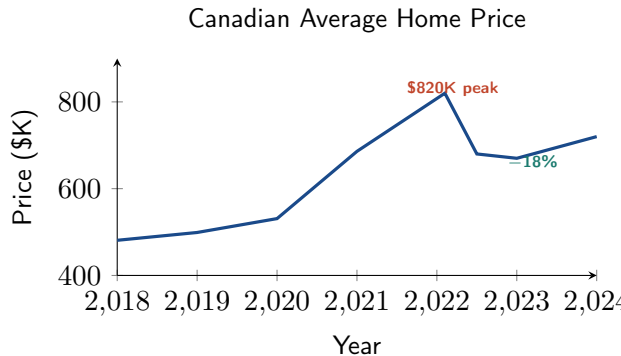


The Numbers

- NASDAQ +527%: 1995 → 2000; then −77%.
- Pets.com: IPO at \$11 → delisted at \$0.19 (9 months).
- Amazon: \$107 → \$5.97 (2001) before recovering.

Drivers: Extrapolation of past returns ·
“Tech will change everything” narrative ·
Social proof · Overconfidence · Availability bias.

The Canadian Housing Bubble (2020–2022)



The Behavioural Story

- BoC cut rates to 0.25% (2020) $\Rightarrow$ prices +64% in 2 years.
- **FOMO**: bought above asking, no conditions.
- **Availability + Extrapolation**: 20 years of rising prices.

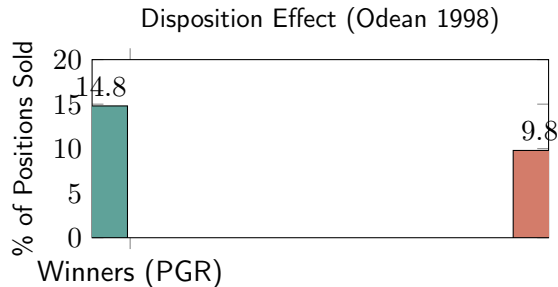
The Crash

BoC raised rates to 5.0% (2022–23). Prices fell **18%**. 300K variable-rate holders in payment shock. Loss aversion made sellers resist cuts.

The Disposition Effect in Markets

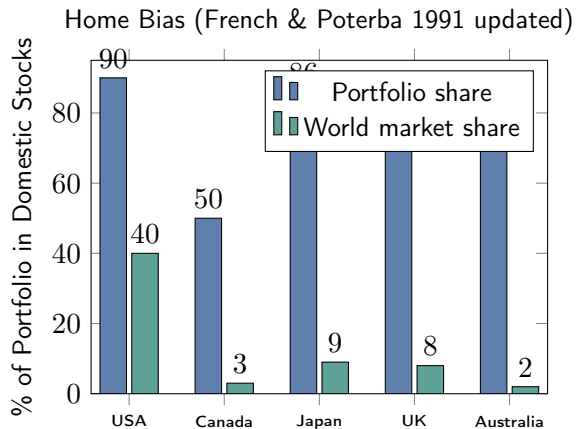
Odean (1998) — US Brokerage Accounts

Investors are 52% more likely to sell winning positions than losing positions. The reference point is the purchase price.



Economic Cost

- Stocks sold (winners): earn **+3.4%** in next year.
- Stocks held (losers): earn **-1.1%** in next year.
- Net annual cost: $\approx$ **4%** of portfolio value.
- Explains momentum anomaly: selling winners $\Rightarrow$ underreaction $\Rightarrow$ continued drift.



The Puzzle

Canadian stocks = **3%** of world market cap.
Canadian investors hold **50%** domestic stocks. This is massive over-concentration.
Optimal diversification says: hold the world portfolio.

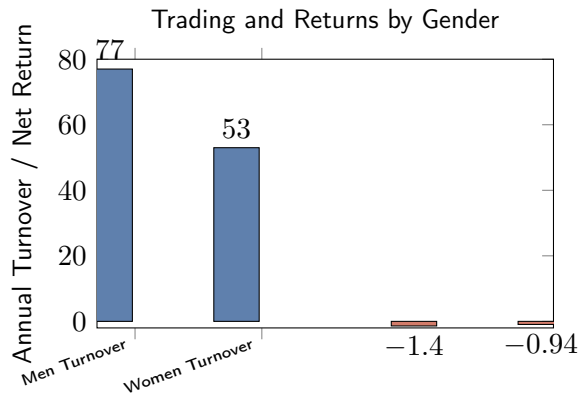
Behavioural Explanations

- **Familiarity:** invest in what you know (Huberman, 2001).
- **Availability:** Canadian stocks more salient.
- **Overconfidence + Herding** by fund managers.

Overconfidence and Excess Trading

Barber & Odean (2001) — “Boys Will Be Boys”

Overconfident investors trade more. Overconfidence is higher in men (psychological research).
Prediction: men trade more than women and earn less.



The Numbers

- Men's annual turnover: **77%**
- Women's annual turnover: **53%**
- Men's net annual return: **-1.4%** vs buy-and-hold
- Women's net annual return: **-0.94%**
- Gap: **0.46% per year** attributable to overconfidence-driven excess trading

Self-Assessment (Anonymous)

1. Ever held a loser to avoid “locking in a loss”? $\Rightarrow$ Disposition Effect
2. Does your family hold $>3\%$ Canadian stocks? $\Rightarrow$ Home Bias
3. Check balances more when markets fall? $\Rightarrow$ Myopic Loss Aversion
4. Changed investment plan after a major market move? $\Rightarrow$ Availability Bias

All four are documented in professional money managers and Nobel Prize winners. Not a failure of intelligence — a predictable consequence of cognitive architecture.

Multiple Biases Combined

1. **Overconfidence:** “Housing prices can only go up” narrative, widely believed.
2. **Availability bias:** 20 years of rising US house prices in memory; no national crash in modern data.
3. **Herding:** All major banks held similar MBS exposures simultaneously.
4. **Confirmation bias:** Rating agency internal models assumed low default correlations (confirming AAA ratings).
5. **Complexity and cognitive overload:** CDO-squared, synthetic CDOs — opaque to even sophisticated investors.
6. **Agency problems:** Mortgage originators bore no default risk.
7. **Extrapolation:** Analysts projected historical growth rates into mortgage default models.

Canada's Relative Success

Canada avoided the worst of 2008 due to: strict mortgage underwriting, mandatory CMHC insurance on high-ratio mortgages, no equivalent subprime market. Canadian unemployment peaked at 8.7% vs US 10%.

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Discussion Questions

- Q1. “Should retail investors just buy index funds? Is that the behavioural economics prescription?”
- Q2. “If you were running a \$1 billion investment fund, what behavioural guardrails would you put in place to prevent your team from falling victim to disposition, herding, and overconfidence?”
- Q3. “Is it irresponsible for the Bank of Canada to set interest rates knowing that low rates fuel asset bubbles? Should the BoC consider asset prices in its mandate?”

Key Takeaways — Lecture 8

- ✓ **EMH:** compelling benchmark — 91% of active Canadian managers underperform. But systematic anomalies exist.
- ✓ **Noise trader model:** rational arbitrageurs can't fully correct mispricing; noise traders can earn higher returns by bearing sentiment risk.
- ✓ **Limits of arbitrage:** fundamental risk, noise trader risk, synchronisation risk, agency costs.
- ✓ **Bubbles:** extrapolation + narrative + herding + overconfidence $\Rightarrow$ prices decouple from fundamentals.
- ✓ **Disposition effect:** sell winners too early, hold losers too long. Costs $\approx 4\%$ /year.
- ✓ **Home bias:** Canadians hold 50% domestic despite Canada = 3% of world market cap.
- ✓ **2008:** overconfidence + herding + extrapolation + complexity $\Rightarrow$ systemic failure.

Next: Lecture 9: Nudges & Choice Architecture — We use everything we've learned to improve decisions *without* restricting freedom.